



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. ECE-‘B’

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit II / Operators

Course Outcome: CO 2

Topic Learning Outcome: TLO 5

Activity Chosen: Virtual Laboratory

Justification:

- The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is an onscreen simulator or calculator that learners use to test ideas and observe results. They allow the learner to “tinker” with laboratory equipment that behaves in almost the same way as it would in a real environment. Learners perform a series of experiments that yield authentic results.
- **Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes. Instructor gave a program to the students.
- The following link is used to access the virtual lab. <https://python-iitk.vlabs.ac.in/> URL for Virtual lab
- Choose Computer Science and Engineering. This page contains broad area of virtual labs
- Select Python Programming Lab from the listed.
- Choose List of Experiments from the drop down list.
- Select arithmetic operations and the students were asked to answer the pretest.
- Procedure for steps for simulator is explained clearly.
- Demonstrate the operations using simulation button.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1
CO 4	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Students can able to apply logical thinking and make the student able to think about problem solving approach	Students will be able to approach the problem by applying the types of operators they learnt and solve the simple and complex engineering problems	Students will be able to interpret the complex engineering problems using the precedence of operators.	The problem solving skill earned through this activity helps the students in solving various problems in information and communication engineering.

- **Images / Screenshot of the practice:**

Computer Science and Engineering > Python Programming Lab > Experiments

Arithmetic Operations

Steps of simulator :

1. Read the simulator details.
2. Enter the values you want to operate.
3. Select the desired type of operation from the operations list.
4. Press CALCULATE to proceed.
5. Press NEXT to see the execution of the code.
6. Relevant line in the code will be highlighted.
7. The local variables will be shown in the Output Panel with their values.

Fig 1: Demonstration on operators

Analysis of Arithmetic Operators

Problem?	Step Execute	Result
Division 34 56 Start Next Reset	<pre>a = Input('Enter a Number') b = Input('Enter a Number') c = a / b print('The Division is :',c)</pre>	a: 34 b: 56 c: 0.6071428571428571 Output: The Division is : 0.6071428571428571

Fig:2 Demonstration on analysis of arithmetic operators

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- This helped the students to understand the basics of Arithmetic Operations used in Python Programming Language with the help of an iterative simulator.
 - Students said that the pretest and posttest is very useful to recollect the concepts.

- ❖ **Benefit of the practice:**

- Based on the activity student's felt that it was easy to understand the concept.
- Most of the students were actively participated and enjoyed once they answered correctly in pretest.

❖ ***Challenges faced in implementation:***

- Some students only involved in the demonstration activity.
- The same activity will be planned in the laboratory session to make all the students familiar in the virtual environmaent.

References:

1. <https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/index.html>
2. <https://www.ritrjpm.ac.in/images/computer-science/Virtual%20Lab-Routing%20in%20MANET.pdf>
3. https://www.ritrjpm.ac.in/images/computer-science/40.EC8393_virtual%20lab.pdf

Signature of Faculty Member

HOD